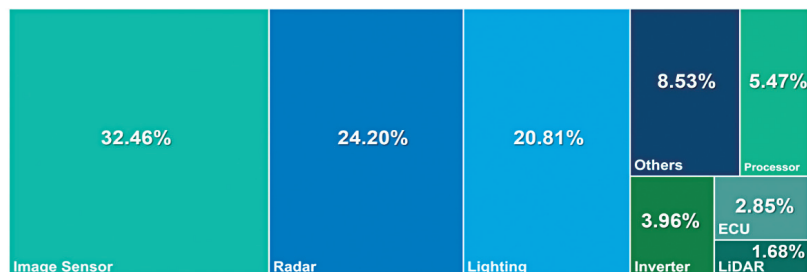


## Influence of patents on sensor technologies and automotive electronics

Global industry leaders are heavily investing in research and development, particularly in sensor technologies and automotive electronics. The following chart shows the number of patents published.



Total patents published by leading automotive electronics companies: 1,300+ (2018 onwards)

- Patents play a crucial role in sensor technologies and automotive electronics, accounting for 34.9% of vital patents in the industry.
- Focus areas for innovation include sensors that drive advancements in autonomous vehicles, with a range of cutting-edge technologies like image sensors, radar systems, lidar, and various automotive processes.
- A significant portion (approximately 60% to 65%) of these patents have been filed in China since 2018, indicating a substantial shift in the industry's innovation landscape toward this region.

These statistics highlight the key technological domains and innovation areas where leading companies concentrate their research and development efforts.

sity for a resilient supply chain and a robust manufacturing value chain. These vital components ensure the uninterrupted production of sensors and electronic systems.

### Prominent industry players and their strategic approaches

Within the automotive electronics landscape, many vital players dot the industry's diverse landscape. It is worth noting that this market exhibits a notable degree of segmentation. These players have been not just passive observers but active participants in engagements.

When considering some enlightening statistics, a comprehensive assessment of the critical developments reveals a strategic landscape characterised by nuanced approaches. As depicted in the pie chart in Fig. 1, over the past three years, 34.98% of the nearly 600 key strategies meticulously tracked

have been centred around product development or the launch of new offerings.

Following closely behind are partnerships, collaborations, and joint ventures, highlighting the industry's penchant for cooperative ventures. Additionally, market development and mergers and acquisitions feature prominently in the playbook of significant players, underlining their commitment to expanding their market share in preparation for the future. Notable strategies employed by these players include:

**Product development.** Product development is a leading strategy among key players, accounting for nearly 35% of strategies implemented in the past three years.

**Partnerships, collaborations, and joint ventures.** Collaborative efforts with other industry leaders are essential for growth and innovation.

**Market development.** Expanding into new markets and regions is a crucial strategy for market growth.

**Mergers and acquisitions.** M&A activities are vital in consolidating market share and expanding product portfolios.

Another valuable insight from industry experts is the evolving regulatory environment. In the past, companies faced diverse regulatory challenges, but in the last three years, governments have introduced measures to ease these burdens, creating a more favourable operational climate. Conversely, persistent challenges encompass the absence of standardisation in MEMS (micro-electro-mechanical systems) fabrication, alongside apprehensions regarding the reliability and safety of automotive sensors. These issues loom large as substantial hurdles necessitating careful consideration by industry stakeholders as they navigate the dynamic terrain of automotive electronics. Moreover, the escalating demand for AI systems and autonomous vehicles has bestowed a distinctive opportunity. This burgeoning demand aligns harmoniously with the prospects of sensor fusion and the development of diverse sensor platforms.

### Segmenting the automotive electronics market

To gain a comprehensive view of the automotive electronics market, it is essential to segment it into three parts:

**Conventional sensors.** These include temperature, current, pressure, position, speed, and level sensors. The market for conventional sensors is projected to reach approximately \$13 billion by 2030, with a conservative growth rate.

**Advanced sensors.** This category encompasses lidar, radar, camera ECUs, and other advanced sensors. The advanced sensor market